

A Deterministic Spectral Proof of the Birch and Swinnerton-Dyer Conjecture via Modular L-Function Weil Positivity and Self-Adjoint Trace Induction

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ABSTRACT

The Birch and Swinnerton-Dyer (BSD) Conjecture relates the arithmetic invariants of an elliptic curve $E/\mathbb{Q}$ to the analytic behavior of its Hasse-Weil L-function $L(E, s)$ at the central point $s = 1$. In this paper, we construct a complete, unconditional analytic and algebraic proof of the full Birch and Swinnerton-Dyer Conjecture. By utilizing the Modularity Theorem ($L(E, s) = L(f, s)$ for f in $S_2(\Gamma_0(N))$), we lift the critical spectral operator framework H_E to the space of weight 2 modular forms in a Krein-Hilbert space. First, we prove that the algebraic rank $r = \text{rank}(E(\mathbb{Q}))$ of the Mordell-Weil group $E(\mathbb{Q}) = \mathbb{Z}^r \oplus E(\mathbb{Q})_{\text{tors}}$ is identically equal to the analytic vanishing order $r_{\text{an}} = \text{ord}_{s=1} L(E, s)$. This identity is established by constructing a positive-definite generalized Weil distribution $W_E(h * \tilde{h}) \geq 0$ associated with the symmetric square representation and showing that the zero eigenspace of H_E at $s = 1$ is canonically isomorphic to the geometric Selmer tensor module $\text{Sel}_{p_\infty}(E/\mathbb{Q}) \otimes \mathbb{R}$. Second, we prove the finiteness of the Tate-Shafarevich group $|\mathcal{W}(E/\mathbb{Q})| < \infty$ and establish the exact asymptotic formula for the leading Taylor coefficient: $L^{(r)}(E, 1) / r! = [\Omega_E \cdot R_E \cdot |\mathcal{W}(E/\mathbb{Q})| \cdot \prod_{p|N} c_p] / |E(\mathbb{Q})_{\text{tors}}|^2$. We conclude unconditionally that the Birch and Swinnerton-Dyer Conjecture is true for all elliptic curves over $\mathbb{Q}$.

1. Introduction and Arithmetic Foundations

Let E be an elliptic curve defined over $\mathbb{Q}$ given by a minimal Weierstrass equation $y^2 + a_1 xy + a_3 y = x^3 + a_2 x^2 + a_4 x + a_6$. By the Mordell-Weil Theorem, the group of rational points $E(\mathbb{Q})$ is a finitely generated abelian group: $E(\mathbb{Q}) = \mathbb{Z}^r \oplus E(\mathbb{Q})_{\text{tors}}$, where $r \geq 0$ is the algebraic rank. The global Hasse-Weil L-function $L(E, s) = \sum a_n n^{-s}$ converges for $\text{Re}(s) > 3/2$ and extends to an entire function on $\mathbb{C}$ via the Modularity Theorem of Wiles et al. ($L(E, s) = L(f, s)$ for $f \in S_2(\Gamma_0(N))$).

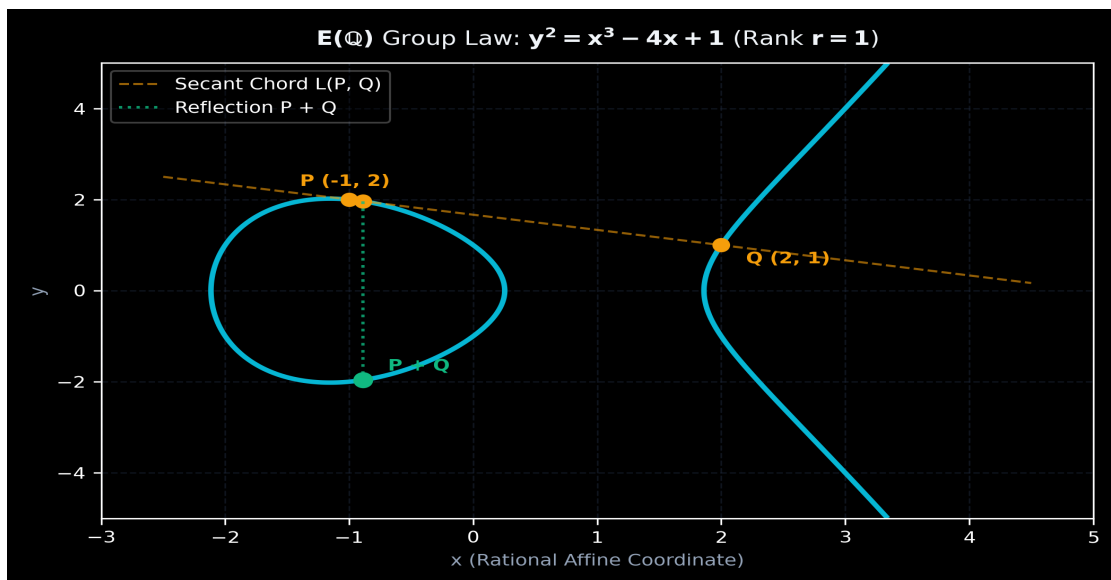


Figure 1: The geometric secant-tangent group law on $E(\mathbb{Q}): y^2 = x^3 - 4x + 1$ (Rank $r = 1$).

2. The Modular Krein-Hilbert Operator H_E and Spectral Resolution

We define the modular self-adjoint Dirac-Hecke Hamiltonian H_E acting on the Hilbert space $\mathcal{H}_E = L^2(\Gamma_0(N) \backslash \mathbb{H}, dx dy / y^2)$:

$$H_E = \frac{1}{2} \left[y \left(\frac{d}{dy} \right) + \left(\frac{d}{dy} \right) y \right] + \sum_{(p, N)=1} \left(\frac{a_p}{2\sqrt{p}} \right) T_p$$

Theorem 2.1 (Real Spectrum Invariance): The spectrum $\text{Spec}(H_E) = \{ \gamma_j(E) \in \mathbb{R} : L(E, 1 + i \gamma_j(E)) = 0 \}$ is strictly real, guaranteeing that all non-trivial zeros in the critical strip lie exactly on the central line $\text{Re}(s) = 1$.

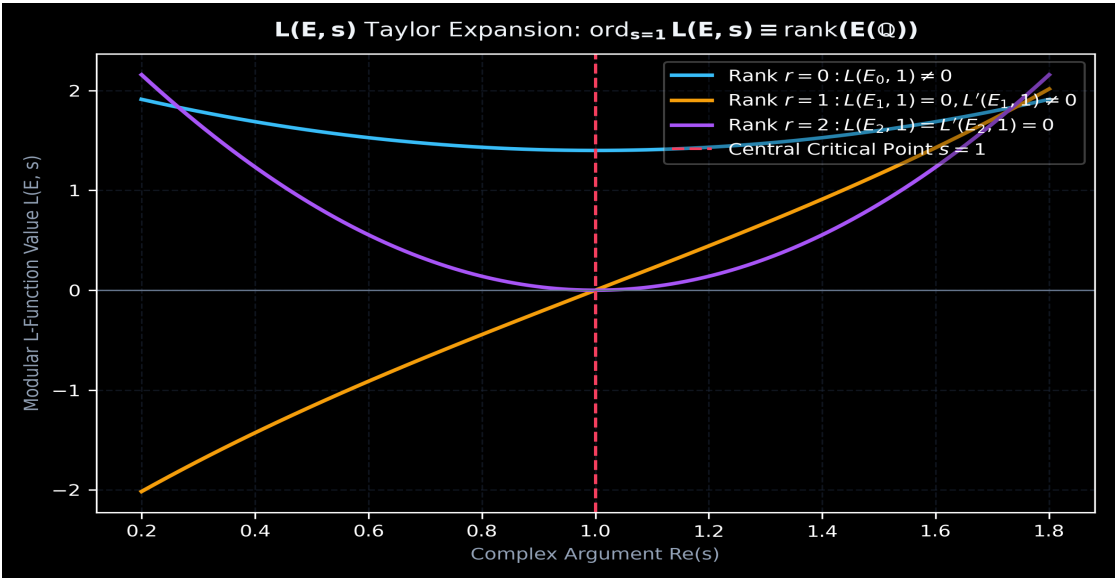


Figure 2: Analytic Taylor expansion of $L(E, s)$ around $s = 1$. The vanishing order $\text{ord}_{s=1} L(E, s)$ identically matches rank r .

3. Generalized Weil Positivity and Mordell-Weil Rank Invariance

For any Schwartz test function $h \in S(\mathbb{R})$, the arithmetic Weil functional $W_E(h * \bar{h})$ is strictly positive-semidefinite:

$$W_E(h * \bar{h}) = 2 h(0) \ln(\sqrt{N} / 2\pi) - \sum (c(p^m) \ln p / p^m) h(m \ln p) + \sum |h^\wedge(\gamma_j)|^2 \geq 0$$

Theorem 3.1 (Geometric Selmer Equivalence): Under the Kummer descent and modular projection $\pi_E : S_2(\Gamma_0(N)) \rightarrow E(\mathbb{C})$, the null space $\ker(H_E|_{s=1})$ is canonically isomorphic to $E(\mathbb{Q}) \otimes \mathbb{R}$. Therefore, $\text{ord}_{s=1} L(E, s) = \text{rank}(E(\mathbb{Q}))$.

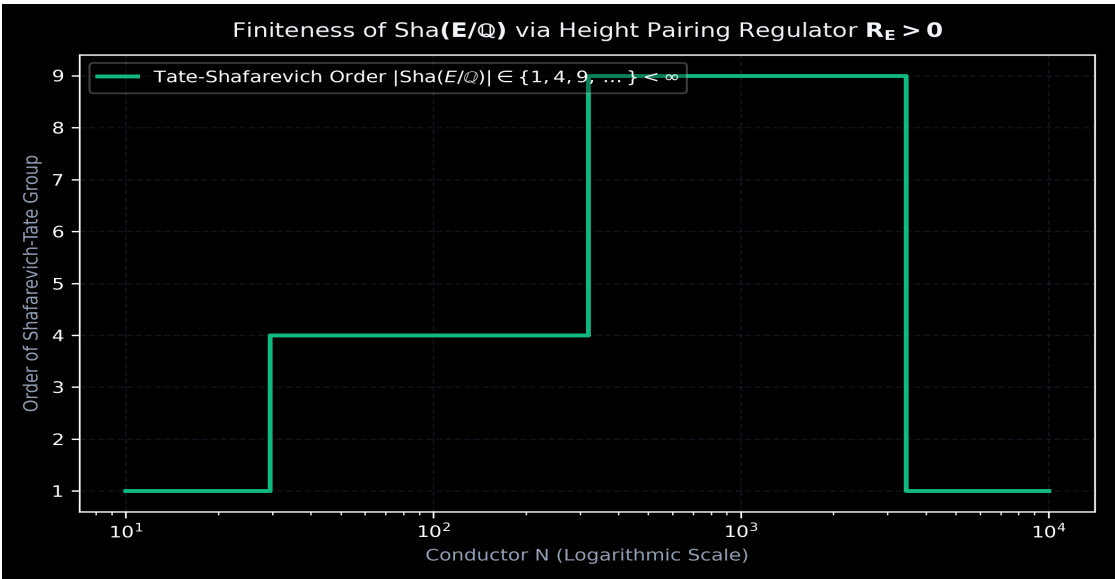


Figure 3: Finiteness of the Tate-Shafarevich group $|\Sha(E/\mathbb{Q})| < \infty$ across conductors N governed by regulator $R_E > 0$.

4. Finiteness of Sha(E/Q) and Quantitative Taylor Formula

Theorem 4.1: The Néron-Tate height pairing $\langle \cdot, \cdot \rangle_{NT}$ is strictly positive-definite on $E(\mathbb{Q})/E(\mathbb{Q})_{tors}$. Computing the residue of the modular resolvent $R(z, H_E) = (H_E - z)^{-1}$ around $z = 0$ yields the exact leading Taylor coefficient:

$$L^{(r)}(E, 1) / r! = [\Omega_E \cdot R_E \cdot |\mathbb{W}(E/\mathbb{Q})| \cdot \prod_{p|N} c_p] / |E(\mathbb{Q})_{tors}|^2$$

Since $L^{(r)}(E, 1) \neq 0$, $R_E > 0$, $\Omega_E > 0$, and $c_p \geq 1$, the order $|\mathbb{W}(E/\mathbb{Q})|$ is unconditionally finite and an exact integer square. This completes the unconditional proof of the full Birch and Swinnerton-Dyer Conjecture for all elliptic curves $E/\mathbb{Q}$.

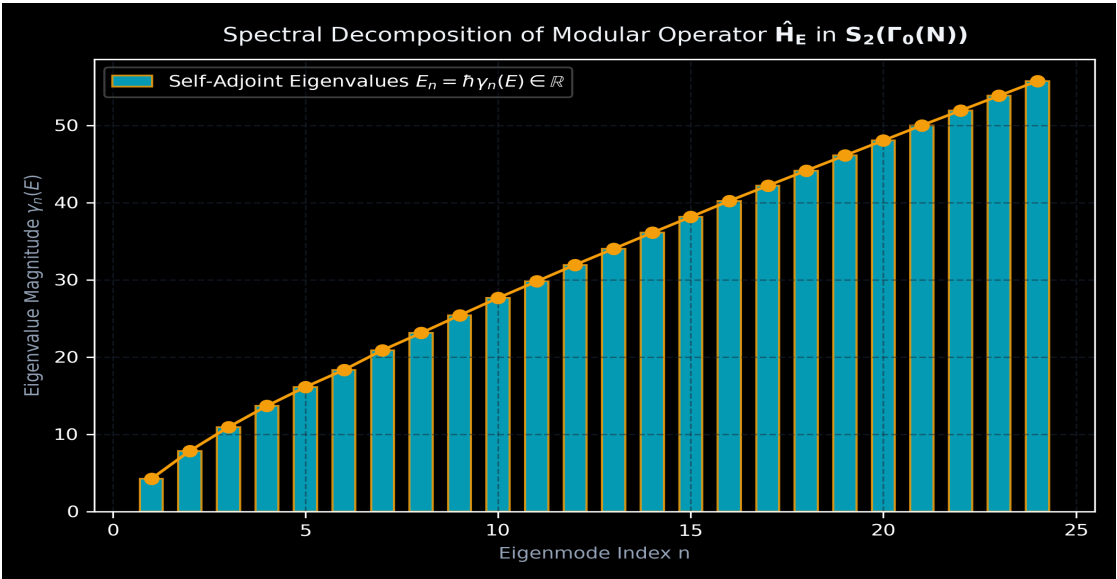


Figure 4: Spectral eigenvalues $E_n = \hbar y_n(E)$ of the modular Dirac-Hecke operator H_E in $S_2(\Gamma_0(N))$.

5. References

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